

AI-based Multimedia Data Compression: Perception Utility Optimization and Standardization

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Abstract

Different 2D and 3D visual data have been widely used in applications such as UHDTV, mobile phones, autonomous driving, robots, and UAVs, etc. The large-scale data amount has raised the research and development trends of image and video coding, and point cloud coding. Compression technologies can efficiently relieve the burden of communication and storage. This tutorial will discuss diverse aspects of multimedia data compression, including datasets, perception models, deep learning-based coding methods, AI-based standards, open source projects, and future research.

CCS Concepts

• Computing methodologies → Image compression; Image representations; • Mathematics of computing → Coding theory; • Information systems → Multimedia streaming.

Keywords

Multimedia Data Compression, Image and Video Coding, 3D Point Cloud Coding, Perception-Driven Coding, AI-Based Standards

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1 Context and Objectives

The realm of multimedia data, spanning 2D images and videos [5, 17], and 3D point clouds [6, 9, 19], has witnessed explosive growth and caused a bottleneck for storage and transmission, which impels the development of data compression technologies.

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Traditional image and video coding methods depend on hand-crafted tools [1] and suffer from the optimization dilemma. Deep learning has revitalized the compression research [5]. For image coding, learning methods can easily exploit channel-modality redundancy [46], and dynamic transform routing to balance computational efficiency and visual quality [32, 43]. For video coding, artifact reduction [25] can effectively improve the fidelity. Subjective quality oriented rate control [39] and saliency-driven bit allocation [22] further optimize human and machine perception. OpenDIC [18], OpenDMC [11], OpenMVC [47], and SCID-Compress900 [49] can offer comprehensive evaluation platforms and datasets.

Point clouds render the compression far more complex than that of 2D visual media [27, 45]. Early efforts focused on non-learning paradigms, exemplified by G-PCC and V-PCC standards [1, 16, 42]. Deep learning has emerged as a transformative force in advancing point cloud coding [12]. Innovations such as octree-based context modeling [4], non-rigid geometry coding [28], diffusion-based and graph-based attribute coding [29, 30] have significantly improved the coding performance. Region-of-interest (ROI) guidance [37], and view-guided learned compression [48] have optimized human and machine vision tasks [38]. For dynamic point clouds, AdaDPCC [44] jointly optimizes rate, distortion, and complexity.

Standardization organizations are actively integrating artificial intelligence into coding frameworks, e.g., the AVS point cloud group [8] prioritize learning-based solutions [34]. Complementing these advancements, open-source libraries, including OpenPointCloud [14], LearningPCC [35], and PCHMVision [36], provide sources for benchmarking and deployment. For video coding, OpenFastVC [40] and OpenHardwareVC [15] enable efficient implementations, while OpenSEP [41], screen-based subjective tools [2], and VE-bench [31] benefit quality assessment research. DPCSet [13] and T23D-QA [21] provide 3D perception datasets.

Modern compression can jointly optimize human and machine perception. Quality assessment [20, 33] is essential for understanding human perception. Saliency detection [3, 7, 10] and just noticeable distortion (JND) can guide bit allocation. For machine perception, compression must preserve critical features to downstream tasks [37]. Enhancement can complement compression [23, 24, 26].

2 Tutorial Overview

This tutorial covers different aspects of deep learning-based coding, and audiences can grasp the knowledge and principles of this research field, as well as the standardization efforts, open source projects and research trends.

- I. Introduction to Datasets, Trends, and Evaluations

- II. Deep Learning-Based Coding Methods
- III. Deep Learning-Based Coding Standards
- IV. Implementations and Open Source Projects
- V. Conclusion and Discussions for Future Directions

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